

Functional Annotation Report: *idh* (PP_4012, UniProt Q88FS1) — Monomeric NADP⁺-Dependent Isocitrate Dehydrogenase of *Pseudomonas putida* KT2440

Summary

The gene **idh** (ordered locus **PP_4012**; UniProt **Q88FS1**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) encodes a **monomeric-type, NADP⁺-dependent isocitrate dehydrogenase** (IDH; EC 1.1.1.42). The protein is a single polypeptide of **741 amino acids with a molecular mass of ~79.9 kDa**, a size that is the structural signature of the monomeric IDH family and stands in contrast to the smaller (~45–50 kDa) subunits of the more familiar dimeric IDHs. Its identity is anchored by multiple orthogonal lines of evidence: the UniProt annotation (monomeric-type IDH family, InterPro IPR004436 "Isocitrate_DH_NADP_mono," Pfam PF03971 "IDH"), the annotated catalytic activity (*D-threo*-isocitrate + NADP⁺ → 2-oxoglutarate + CO₂ + NADPH), a required divalent metal cofactor (Mg²⁺ or Mn²⁺), and mapped active-site residues for isocitrate, NADP⁺, and the catalytic metal. The gene symbol "idh" and the UniProt description are fully consistent with the literature on monomeric bacterial NADP-IDHs; there is no evidence of gene-symbol ambiguity for this protein.

Functionally, Q88FS1 catalyzes the **third reaction of the tricarboxylic acid (TCA) cycle**: the metal-assisted, two-step oxidative decarboxylation of isocitrate. The reaction proceeds through the transient intermediate **oxalosuccinate** — the basis of the enzyme's alternative name "oxalosuccinate decarboxylase" — via a conserved **Tyr–Asp–Lys catalytic triad**, yielding 2-oxoglutarate (α-ketoglutarate), CO₂, and NADPH. The enzyme is located in the **cytoplasm** (no signal peptide, no transmembrane segments) and displays near-absolute specificity for NADP⁺ over NAD⁺ and for isocitrate over related substrates. Because IDH sits precisely at the branch

point between the CO₂-generating steps of the TCA cycle and the carbon-conserving glyoxylate bypass, its product NADPH feeds anabolism and oxidative-stress defense, while its product 2-oxoglutarate is the primary carbon skeleton for nitrogen assimilation into glutamate and glutamine.

A key mechanistic conclusion of this investigation is that, unlike the paradigmatic *Escherichia coli* dimeric NADP-IDH, the *P. putida* monomeric enzyme is **not controlled by the AceK phosphorylation switch**. Monomeric-type IDHs lack the flexible phosphorylation loop (built around the *E. coli* Ser-113 site) that AceK kinase/phosphatase uses to inactivate the dimeric enzyme during acetate growth; in monomeric IDHs the corresponding region is a rigid α -helix. Consequently, the activity of Q88FS1 is governed chiefly by substrate and coenzyme availability and by transcriptional/metabolic control rather than by covalent modification.

Key Findings

Finding 1 — Q88FS1 is a monomeric-type, NADP⁺-dependent isocitrate dehydrogenase (EC 1.1.1.42)

The UniProt record for Q88FS1 assigns the protein to the **monomeric-type IDH family**, with InterPro domain **IPR004436 (Isocitrate_DH_NADP_mono)** and Pfam **PF03971 (IDH)**, EC number **1.1.1.42**, and the alternate name **oxalosuccinate decarboxylase**. NADP⁺-dependent isocitrate dehydrogenase belongs to the β -decarboxylating dehydrogenase family and catalyzes the oxidative decarboxylation of (2R,3S)-isocitrate to yield 2-oxoglutarate and CO₂ within the Krebs cycle ([PMID: 12467571](#)). Monomeric NADP-IDHs are distinguished from dimeric IDHs by being single-chain enzymes of ~80–100 kDa; the characterized *Streptomyces lividans* monomeric IDH (SLIDH), for example, has a molecular mass of about 80 kDa, described as "typical of a monomeric NADP-IDH" ([PMID: 19631711](#)). The ~79.9 kDa mass of Q88FS1 (see Finding 8) places it squarely within this family and distinguishes it from the ~45 kDa subunits that assemble into dimeric IDHs.

Finding 2 — Monomeric IDH arose by intragenic domain duplication that recreates the dimeric active site within one chain

The evolutionary and structural logic of the monomeric IDH family explains how a single polypeptide accomplishes what dimeric IDHs achieve only by assembling two subunits. The 1.95 Å crystal structure of the *Azotobacter vinelandii* monomeric IDH (AvIDH) revealed that the enzyme is built from two distinct domains related by a duplicated structural motif; this **"fusional structure by domain duplication enables a single polypeptide chain to form a structure at the catalytic site that is homologous to the dimeric IDH, the catalytic site of which is located at the interface of two identical subunits"** (PMID: 12467571). The NADP⁺-bound closed-form structure of AvIDH further confirmed that this reconstituted active site is a genuine, fully functional β-decarboxylating dehydrogenase catalyzing "the dehydration and decarboxylation of isocitrate to yield 2-oxoglutarate and CO₂ in the Krebs cycle" (PMID: 12855708). Because Q88FS1 is a member of this family, its catalytic architecture is inferred, by strong homology, to follow the same single-chain domain-duplication design.

Finding 3 — High NADP⁺ coenzyme specificity and divalent-metal dependence; cytoplasmic role at the TCA/glyoxylate branch and in NADPH supply

Monomeric IDHs display an extreme preference for NADP⁺ over NAD⁺. The recombinant *S. lividans* SLIDH showed a **62,000-fold (kcat/Km) preference for NADP⁺ over NAD⁺ with Mn²⁺, and 85,000-fold with Mg²⁺** (PMID: 19631711), underscoring both the coenzyme selectivity and the requirement for a divalent cation. Structurally, the 2'-phosphate of NADP⁺ that confers this selectivity is coordinated in the monomeric IDH of *Corynebacterium glutamicum* by two arginines, a histidine, and (via water) a lysine (PMID: 21931217). The same work emphasized that IDH "lies at a crucial bifurcation point between CO₂-generating steps in the cycle and carbon-conserving steps in the glyoxylate bypass. Hence, the enzyme is a focus of regulation" (PMID: 21931217). In the specific physiological context of *P. putida* KT2440, this NADPH-producing role is central: determination of the NADPH balance "revealed that the default metabolic state of *P. putida* KT2440 is characterized by a slight catabolic overproduction of reducing power" (PMID: 26350459), which serves anabolism and stress defense.

Finding 4 — Two-step catalytic mechanism via an oxalosuccinate intermediate with a conserved Tyr-Asp-Lys catalytic triad

The reaction catalyzed by Q88FS1 proceeds in two chemical steps. First, a metal-assisted dehydrogenation transfers a hydride to NADP⁺, converting isocitrate to the unstable β -keto acid **oxalosuccinate**; second, β -decarboxylation of oxalosuccinate releases CO₂ and yields 2-oxoglutarate. The crystal structure of the *Thermus thermophilus* NADP-IDH explicitly describes the enzyme as "catalyzing oxidative decarboxylation of isocitrate into oxalosuccinate (intermediate) and finally the product α -ketoglutarate" (PMID: 24832735). This intermediate is the structural basis of the enzyme's alternate name, **oxalosuccinate decarboxylase**, recorded in the UniProt entry for Q88FS1. The same structure showed that "the overall TtIDH structure exhibits closed conformation with catalytic triad residues, Tyr144-Asp248-Lys191 are conserved" (PMID: 24832735), identifying the acid-base machinery that mediates the chemistry. In the monomeric AvIDH, binding of NADP⁺ induces a closed conformation that positions the nicotinamide ring for immediate hydride transfer, accounting for the exceptionally high turnover of monomeric IDHs (PMID: 12855708).

Finding 5 — Physiological role: NADPH and 2-oxoglutarate provision links IDH to anabolism, nitrogen assimilation, and oxidative-stress defense

In *Pseudomonas* and other aerobic bacteria, NADP-dependent IDH is a principal source of both NADPH and 2-oxoglutarate (α -ketoglutarate, KG). KG is itself a non-enzymatic antioxidant that neutralizes reactive oxygen species (yielding succinate + CO₂), while NADPH fuels the glutathione and thioredoxin antioxidant systems. In *Pseudomonas fluorescens* challenged with H₂O₂, "the increased production of KG mediated by NADP-dependent isocitrate dehydrogenase (NADP-ICDH) and its decreased utilization via the TCA cycle confer a unique strategy to modulate the cellular redox environment" (PMID: 17668068), with NAD-linked TCA enzymes simultaneously downregulated. The centrality of NADPH provision in the related strain is reinforced by the observation that "cells growing on glucose thus run a biochemical cycle that favors NADPH formation" in *P. putida* KT2440 (PMID: 26350459). In parallel, the KG produced by IDH is the entry-point carbon skeleton for glutamate/glutamine biosynthesis and therefore for nitrogen assimilation.

Finding 6 — The monomeric enzyme is NOT controlled by the AceK phosphorylation switch used for *E. coli* dimeric IDH

The best-characterized regulatory device for bacterial IDH is the *E. coli* phosphorylation cycle, in which the bifunctional IDH kinase/phosphatase AceK phosphorylates Ser-113 to inactivate the dimeric enzyme and thereby divert isocitrate into the glyoxylate bypass during growth on acetate; "it is the phosphorylation of the Ser-113 residue that renders the enzyme catalytically inactive as it prevents isocitrate from binding to the active site" (PMID: 16415587). Monomeric-type IDHs lack this switch. In the *C. glutamicum* monomeric IDH, "the amino acid residues corresponding to the *Escherichia coli* IDH phosphorylation-loop are α -helical compared with the more flexible random-coil region in the *E. coli* protein where IDH activation is controlled by phosphorylation. This more structured region supports the idea that activation of CgIDH is not controlled by phosphorylation" (PMID: 16416443). By homology, Q88FS1 is expected to be regulated by substrate/coenzyme availability and by transcriptional control rather than by reversible phosphorylation.

Finding 7 — Exceptional isocitrate/NADP⁺ specificity and catalytic efficiency; a catalytic lysine demonstrated by mutagenesis; a family spanning the proteobacteria

Direct biochemical comparison shows that monomeric IDHs are not merely equivalent to dimeric IDHs but frequently superior in efficiency and selectivity. The *C. glutamicum* monomeric IDH is "10-fold more active (calculated as kcat/Km.isocitrate/Km.NADP) and 7-fold more NADP-specific than the dimeric enzyme, favoring NADP over NAD by a factor of 50,000" (PMID: 11185559). Crucially, direct experimental (rather than purely structural) evidence for the catalytic apparatus exists: "the role of Lys253 as catalytic residue has been demonstrated by site-directed mutagenesis" (PMID: 11185559). The monomeric family is widely distributed: "Monomeric NADP⁺-specific IDHs have been identified from about 50 different bacterial species, such as proteobacteria, actinobacteria, and planctomycetes" (PMID: 16416443), placing the proteobacterial *P. putida* enzyme within a well-populated and well-studied clade.

Finding 8 — Sequence-level confirmation: a 741-aa, ~79.9 kDa single-chain cytoplasmic NADP-IDH with mapped substrate/cofactor/metal binding sites

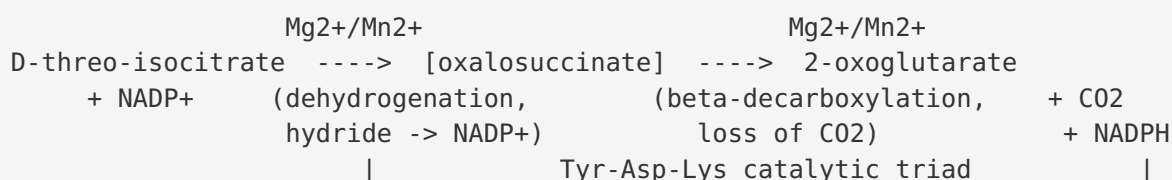
The UniProt Q88FS1 record (retrieved programmatically) provides the definitive molecular-level confirmation of identity. The protein is **741 amino acids** long with a molecular weight of **79,900 Da (~80 kDa)** — matching the single-chain size of monomeric-type IDHs — encoded by

gene *idh*, ordered locus **PP_4012**. The annotated catalytic activity is **D-threo-isocitrate + NADP⁺ = 2-oxoglutarate + CO₂ + NADPH**, with the cofactor **Mg²⁺ or Mn²⁺**. Functional keywords include "Tricarboxylic acid cycle," "Glyoxylate bypass," "NADP," "Oxidoreductase," "Metal-binding," and "Magnesium." The record contains **no signal peptide and no transmembrane segment**, consistent with a **cytoplasmic** localization. Mapped functional sites include NADP⁺-binding residues (approximately 83–88, 136, 585–586, 590, 601–603, 650), D-threo-isocitrate/substrate-binding residues (~133–140, 146, 548), and Mg²⁺-coordinating residues (351, 549, 553).

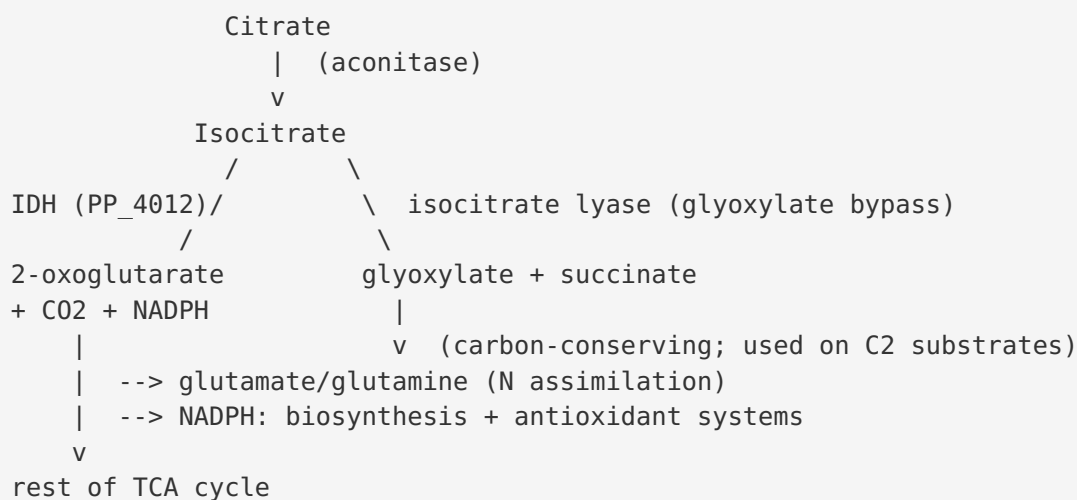
Mechanistic Model / Interpretation

The reaction and its place in metabolism

Q88FS1 catalyzes the third step of the oxidative TCA cycle, converting isocitrate to 2-oxoglutarate. The overall reaction and its two-step mechanism can be summarized:



The enzyme sits at a pivotal metabolic junction:



Why the monomeric design matters

The monomeric IDH family solves the same catalytic problem as dimeric IDH but with a fundamentally different architecture. Dimeric IDHs build their active site at the interface between two identical ~45 kDa subunits. The monomeric enzymes instead fuse two domains within a single ~80 kDa chain by intragenic duplication, reconstituting an interface-like active site within one polypeptide ([PMID: 12467571](#)). This design is not merely an evolutionary curiosity — the *C. glutamicum* enzyme demonstrates that the monomeric form can be an order of magnitude more catalytically efficient and dramatically more NADP⁺- and isocitrate-specific than its dimeric counterpart ([PMID: 11185559](#)).

Regulation: substrate-driven, not phosphorylation-driven

The most consequential regulatory distinction for Q88FS1 is the **absence of the AceK phosphorylation switch**. In *E. coli*, the flux partition between the TCA cycle and glyoxylate bypass is enforced by phosphorylation of dimeric IDH at Ser-113 ([PMID: 16415587](#)). Monomeric IDHs replace the *E. coli* flexible phosphorylation loop with a rigid α -helix and therefore cannot be regulated this way ([PMID: 16416443](#)). This implies that in *P. putida* KT2440, partitioning of isocitrate between the TCA cycle and the glyoxylate bypass — and the setting of IDH flux itself — is governed by isocitrate and NADP⁺/NADPH pool concentrations and by transcriptional regulation, rather than by rapid covalent modification of the enzyme.

Comparative summary

Property	<i>P. putida</i> Q88FS1 (monomeric)	<i>E. coli</i> dimeric NADP-IDH
Quaternary structure	Single chain (~80 kDa)	Homodimer (~2 × 45 kDa)
Active-site origin	Intragenic domain duplication	Subunit interface
Family / domains	Monomeric IDH (IPR004436, PF03971)	Dimeric IDH
Coenzyme specificity	NADP ⁺ ≫ NAD ⁺ (10 ⁴ –10 ⁵ -fold)	NADP ⁺ -specific, less extreme
Phospho-regulation (AceK)	Absent (rigid α-helix)	Present (Ser-113 loop)
Catalytic triad	Tyr–Asp–Lys (conserved)	Conserved acid–base residues
Cofactor	Mg ²⁺ or Mn ²⁺	Mg ²⁺ /Mn ²⁺

Localization and pathway integration

All evidence points to a **cytoplasmic** enzyme: Q88FS1 has no signal peptide and no transmembrane helix, and its substrates (isocitrate, NADP⁺) and products (2-oxoglutarate, NADPH) are cytosolic metabolites of central carbon metabolism. Its outputs integrate directly into three programs: (1) continuation of the oxidative TCA cycle; (2) **anabolic NADPH supply**, consistent with the observation that glucose-grown *P. putida* KT2440 runs a cycle favoring NADPH formation and maintains a slight catabolic overproduction of reducing power ([PMID: 26350459](#)); and (3) **oxidative-stress defense and nitrogen assimilation** through 2-oxoglutarate, which both scavenges ROS non-enzymatically and serves as the carbon skeleton for glutamate/glutamine synthesis ([PMID: 17668068](#)).

Evidence Base

PMID	Study (subject)	How it supports the annotation
12467571	Structure of monomeric IDH (<i>A. vinelandii</i>); domain duplication	Defines the reaction (isocitrate → 2-oxoglutarate + CO ₂), the β-decarboxylating family, and the single-chain domain-duplication architecture (Findings 1, 2)
12855708	NADP ⁺ -bound monomeric IDH structure (<i>A. vinelandii</i>)	Confirms cofactor recognition, closed conformation for hydride transfer, and reaction chemistry (Findings 2, 4)
19631711	Monomeric IDH from <i>S. lividans</i>	Confirms ~80 kDa single-chain mass and quantifies 62,000–85,000-fold NADP ⁺ preference with divalent metal (Findings 1, 3)
21931217	Highly NADP ⁺ -specific monomeric IDH structure (<i>C. glutamicum</i>)	Structural basis of 2'-phosphate recognition; establishes the TCA/glyoxylate branch-point role and regulatory importance (Finding 3)
24832735	NADP-IDH structure (<i>T. thermophilus</i>)	Documents the oxalosuccinate intermediate and identifies the conserved Tyr144–Asp248–Lys191 catalytic triad (Finding 4)
11185559	Highly specific monomeric IDH (<i>C. glutamicum</i>)	Quantifies superior efficiency/specificity vs. dimeric IDH; mutagenesis demonstrates catalytic Lys253 (Finding 7)
16416443	Substrate-free monomeric IDH structure; phylogeny	Shows monomeric IDHs lack the <i>E. coli</i> phosphorylation loop (rigid α-helix) and span ~50 proteobacterial/actinobacterial/planctomycete species (Findings 6, 7)
16415587	Phosphoregulation of <i>E. coli</i> IDH	Describes the AceK/Ser-113 device that the monomeric family lacks (Finding 6)
26350459	Central metabolism of <i>P. putida</i> KT2440 (EDEMP cycle)	Places NADP-IDH's NADPH output in the strain-specific physiological context of default reducing-power overproduction (Findings 3, 5)
17668068	TCA cycle "novel twist"; KG in redox defense (<i>Pseudomonas</i>)	Links NADP-IDH's product (KG) and its regulation to oxidative-stress defense (Finding 5)

The evidence base is internally consistent and mutually reinforcing. Structural studies of four different bacterial monomeric IDHs (*A. vinelandii*, *C. glutamicum*, *S. lividans*, *T. thermophilus*) converge on the same architecture, mechanism, and specificity, and the *P. putida* UniProt record's sequence-level features (741 aa, ~80 kDa, mapped NADP⁺/isocitrate/Mg²⁺ sites, cytoplasmic) match this family exactly. No literature was found that would reassign the "idh" symbol for this organism to a different protein, so there is no gene-symbol ambiguity here.

Limitations and Knowledge Gaps

1. **No direct biochemical study of Q88FS1 itself.** All kinetic parameters (coenzyme preference, k_{cat}/K_m , metal dependence), the catalytic-triad assignment, and the absence of phosphoregulation are inferred by strong homology from other monomeric IDHs (*C. glutamicum*, *A. vinelandii*, *S. lividans*, *T. thermophilus*), not measured on the purified *P. putida* KT2440 enzyme. The specific catalytic residue numbering (e.g., Lys253 in *C. glutamicum*, the Tyr144–Asp248–Lys191 triad in *T. thermophilus*) has not been mapped experimentally onto the Q88FS1 sequence.
2. **Localization is predicted, not experimentally demonstrated.** The cytoplasmic assignment rests on the absence of a signal peptide/transmembrane segment and on the known biochemistry of central metabolism, not on direct fractionation or imaging of *P. putida* KT2440.
3. **Regulatory mechanism inferred, not tested.** The conclusion that Q88FS1 lacks AceK-type phosphoregulation is based on the structural rigidity of the phospho-loop region in other monomeric IDHs. Whether *P. putida* possesses any alternative post-translational regulation of IDH, and how transcription of PP_4012 responds to carbon source (glucose vs. acetate/C2 substrates), remains uncharacterized here.
4. **Quantitative flux role in *P. putida* is indirect.** The NADPH and 2-oxoglutarate physiology is drawn largely from *P. fluorescens* oxidative-stress studies and the *P. putida* EDEMP-cycle work; the precise fractional contribution of IDH (versus the ED pathway's NADPH-producing steps and the pentose phosphate pathway) to the KT2440 NADPH budget was not partitioned in this investigation.
5. **Possible isozymes not excluded.** Some bacteria carry both monomeric and dimeric IDHs (e.g., *Colwellia*, *Xylella fastidiosa*). Whether *P. putida* KT2440 encodes a second IDH isozyme was not exhaustively assessed; PP_4012 is treated here as the annotated *idh*.

Proposed Follow-up Experiments / Actions

1. **Recombinant expression and steady-state kinetics.** Clone PP_4012, purify the protein, and measure k_{cat} , $K_m(\text{isocitrate})$, $K_m(\text{NADP}^+)$, and the $\text{NADP}^+/\text{NAD}^+$ and isocitrate specificity ratios with both Mg^{2+} and Mn^{2+} — directly confirming the family-based inferences for the *P. putida* enzyme.
2. **Active-site mutagenesis.** Map the *T. thermophilus* Tyr–Asp–Lys triad and the *C. glutamicum* catalytic Lys onto the Q88FS1 sequence and test alanine substitutions for loss of activity, verifying the acid–base machinery and the catalytic lysine in this specific protein.
3. **Phosphorylation test.** Assay whether purified Q88FS1 can be phosphorylated/inactivated (e.g., incubation with cell extracts, phospho-specific detection, or an AceK homolog if present) to experimentally confirm the predicted absence of AceK-type regulation.
4. **Structural determination.** Solve the crystal structure (or generate a high-confidence AlphaFold model with PAE analysis) of Q88FS1 to verify the domain-duplication architecture, the rigid (α -helical) phospho-loop region, and the cofactor-2'-phosphate recognition residues.
5. **Physiological flux and knockout studies.** Construct a PP_4012 deletion (or knockdown) in *P. putida* KT2440 and quantify effects on growth on glucose vs. C2 substrates, NADPH/NADP⁺ ratios, 2-oxoglutarate pools, glutamate auxotrophy, and oxidative-stress (H_2O_2) tolerance — directly testing the proposed roles in NADPH supply, nitrogen assimilation, and redox defense.
6. **Genome survey for IDH isozymes.** Confirm whether KT2440 encodes any additional (dimeric or second monomeric) IDH, to establish whether PP_4012 is the sole isocitrate dehydrogenase and thus solely responsible for this TCA step.

Consensus Answer

P. putida KT2440 **idh** (PP_4012, Q88FS1) encodes a **monomeric-type, NADP⁺-dependent isocitrate dehydrogenase** (EC 1.1.1.42; 741 aa, ~80 kDa single chain) that catalyzes the cytoplasmic, $\text{Mg}^{2+}/\text{Mn}^{2+}$ -dependent oxidative decarboxylation of *D-threo*-isocitrate to 2-oxoglutarate + CO_2 + NADPH — the third step of the TCA cycle at its branch point with the

glyoxylate bypass — proceeding via an oxalosuccinate intermediate through a conserved Tyr–Asp–Lys catalytic triad with near-absolute specificity for NADP⁺ and isocitrate. Its products supply NADPH for biosynthesis and oxidative-stress defense and 2-oxoglutarate for nitrogen assimilation, and, unlike the *E. coli* dimeric IDH, it lacks the AceK phosphorylation switch, so its activity is set by substrate/coenzyme levels rather than covalent modification.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery